dash_mlguard v0.3 audit -- fraud detection (temporal + grouped)

dataset: Synthetic transactions, 8,000 rows / 600 users / 90 days | dash_mlguard v0.3.0

FAIL -- critical issues must be resolved before training.

Summary

3	1	0
CRITICAL	WARNING	INFO

Performance: before vs after dash_mlguard

BEFORE (NAIVE)	
accuracy	0.8717
f1	0.6805
roc_auc	0.9065

AFTER (DASH_MLGUARD-CLEANED)	
accuracy	0.8495
f1	0.6569
roc_auc	0.8959

accuracy: 0.8717 -> 0.8495 (-0.0222)
f1: 0.6805 -> 0.6569 (-0.0236)
roc_auc: 0.9065 -> 0.8959 (-0.0106)

Findings

TL011 - Temporal leakage

CRITICAL

What: Test rows include timestamps at or before the latest train timestamp. The model sees the future during training, then evaluates on the past -- production accuracy will be much worse.

Detail: 2400 of 2400 test rows (100.00%) have timestamps at or before the latest train timestamp. train: [2024-01-01 00:00:00 -> 2024-03-31 00:00:00]; test: [2024-01-01 00:00:00 -> 2024-03-31 00:00:00]

Fix: Split chronologically: every test timestamp must be strictly greater than every train timestamp. Use TimeSeriesSplit or sort by time before splitting.

TL012 - Group leakage

CRITICAL

What: The same group identifier (user_id, session_id, patient_id) appears in both train and test splits. Cross-validation overstates accuracy because the model has seen this entity.

Detail: 589 group IDs appear in both train and test (2400 test rows, 100.00% of test). Sample overlapping groups: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9'].

Fix: Group leakage: the same entity (user / patient / session) is in train AND test. Cross-validation will overstate accuracy. Use sklearn.model_selection.GroupKFold or split by the entity ID, not by row.

TL013 - Preprocessing fit-on-full-data leakage

CRITICAL

What: The pipeline produces different transform() outputs for X_test depending on whether it was fit on the training subset or the full dataset. This means scaler / imputer / encoder state depends on test rows -- silent leakage.

Detail: Pipeline output for X_test differs by 0.831279 between (fit on train) and (fit on full data). The pipeline has data-dependent state -- if fit on the full dataset, test information leaks into the model.

Fix: Always fit preprocessing on training data only, then transform(X_test) -- never .fit_transform(X) before splitting. If using sklearn, wrap preprocessing inside a Pipeline and call .fit() on (X_train, y_train) only.

TL006 - Train/test distribution drift

WARNING

What: A feature's distribution differs sharply between train and test (KS>=0.2 numeric, PSI>=0.25 categorical). Your evaluation is measuring transfer, not generalization.

Detail: 1 feature(s) drift between train and test (KS>=0.2 numeric / PSI>=0.25 categorical).

Fix: Train and test should be drawn from the same distribution. If they're not, your evaluation is measuring transfer, not generalization. Re-shuffle, fix temporal leakage, or explicitly model distribution shift.

Cols: timestamp

Generated by dash_mlguard. One call: dash_mlguard.check(X_train, y_train, X_test, y_test)